Lab 7 — Fuzzing for Input Validation Bugs

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Assigned Script: food_delivery.py

Output of Initial Buggy Script

Running the initial buggy application (python food_delivery.py)

```
(venv) C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7>python food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50

(venv) C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7>
```

Hypothesis Test Failures

Fuzz testing failure output using Hypothesis (pytest test_food_delivery.py)

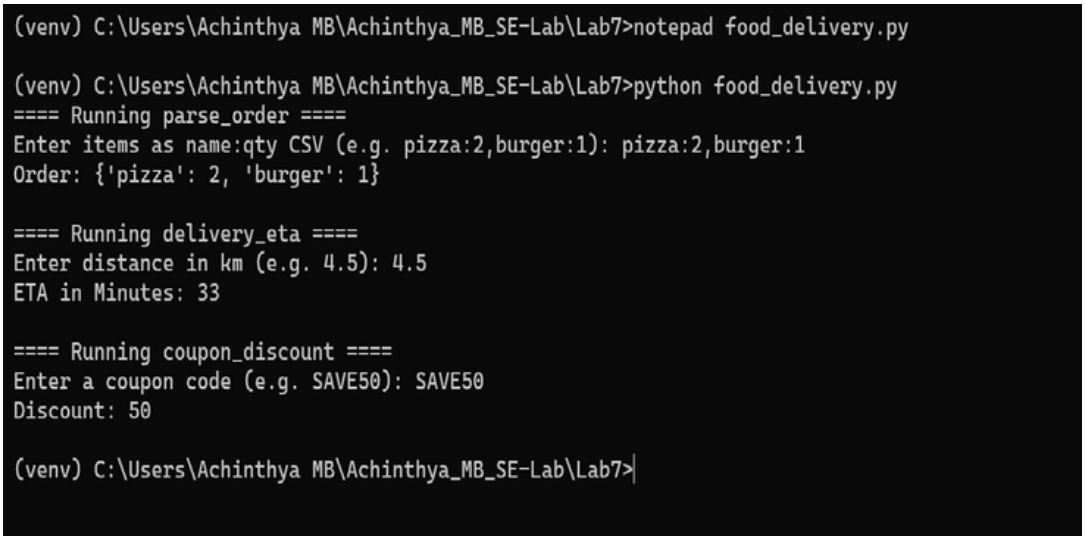
```
    | Traceback (most recent call last):
    |   File "C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7\test_food_delivery.py", line 31, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7\food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'upper'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7\test_food_delivery.py", line 36, in test_coupon_discount_no_crash
    |     pytest.fail(f"coupon_discount crashed unexpectedly with: {type(e).__name__}: {e}")
    |     ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    | Failed: coupon_discount crashed unexpectedly with: AttributeError: 'NoneType' object has no attribute 'upper'
    | Failing test case: test_coupon_discount_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7\test_food_delivery.py:35
    +-----------------------------------
================== short test summary info ==================
FAILED test_food_delivery.py::test_parse_order_no_crash - Failed: parse_order crashed unexpectedly with: AttributeErr...
FAILED test_food_delivery.py::test_delivery_eta_no_crash - Failed: delivery_eta crashed unexpectedly with: OverflowErr...
FAILED test_food_delivery.py::test_coupon_discount_no_crash - BaseExceptionGroup: Hypothesis found 2 distinct failures. (...
==================== 3 failed in 4.46s =====================

(venv) C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7>pytest test_food_delivery.py --tb=short
============================================================================ test session starts ============================
============================================================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Achinthya MB\Achinthya_MB_SE-Lab\Lab7
plugins: hypothesis-6.170.14
collected 3 items

test_food_delivery.py FFF                                                [100%]
```

Output of Fixed Application Script

Running the hardened application (python food_delivery.py)



Re-running Hypothesis Tests on Fixed Code

Passing Hypothesis fuzz tests on hardened code (pytest test_food_delivery.py)

